Data Science and AI for Finance
DA6701

Module – Integrating MPT, CAL, CAPM, and HRP

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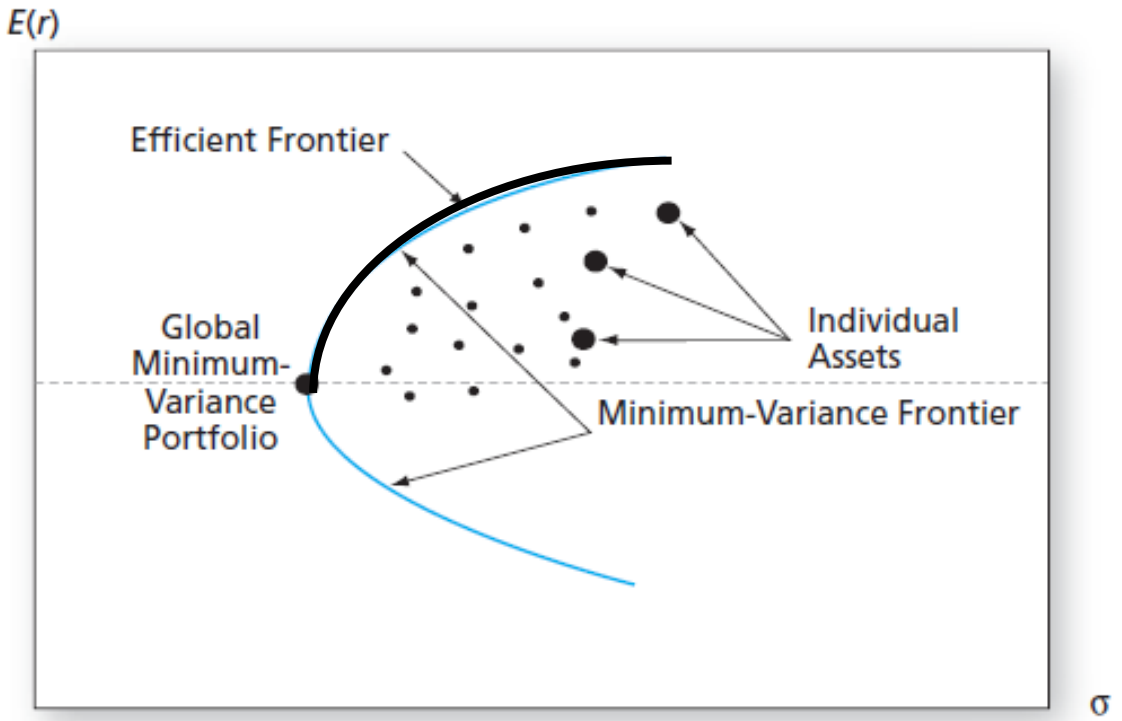
Integrating MPT, CAL, CAPM

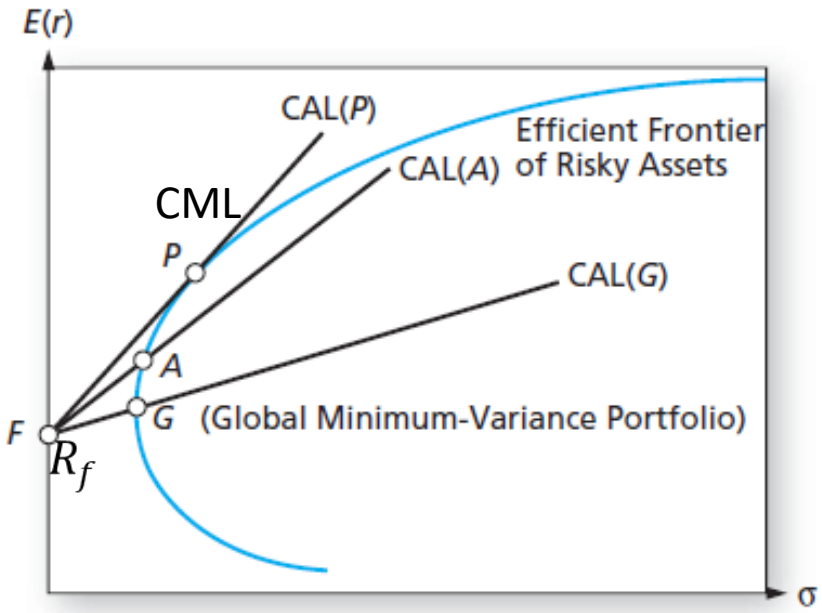
- Markowitz's MPT teaches us that of the many ways we can combine risky assets to create portfolios, there should be an efficient frontier on the risk-reward graph.



Integrating MPT, CAL, CAPM

- CAL teaches us that we can take a risky portfolio (on the efficient frontier) and mix it with a risky free one to create a straight line of options (per portfolio) with the same Sharpe ratio.

- CAPM tells us that there exists a point on the efficient frontier that has the highest Sharpe ratio (tangency portfolio). The line from R_f that is tangent to the point and has the highest slope touching the frontier is the capital market line (CML). The tangency portfolio, according to CAPM, should be the market portfolio, and assets line up on the Security Market Line (SML) (plotted with β as x-axis and $E(R_A)$ as y axis.



Bodie, Zvi, Alex Kane, and Alan Marcus. *Ebook: Essentials of investments: Global edition*. McGraw Hill, 2013.

Nuts and bolts of portfolio selection

- How do we find the efficient frontier?
  - Brute force of exploring weights like a simulation
  - Mathematically:
    - Objective: We seek to minimize risk for a fixed target return.
    - The Formulation: Minimize variance subject to return (μ_p) and budget constraints

$$\min_{w} w^T \Sigma w \quad \text{subject to} \quad w^T \mu = \mu_p, \quad \mathbf{1}^T w = 1$$

    - Lagrangian: We introduce multipliers lambda (return) and gamma (budget).

$$\mathcal{L}(w, \lambda, \gamma) = w^T \Sigma w - \lambda(w^T \mu - \mu_p) - \gamma(\mathbf{1}^T w - 1)$$

Nuts and bolts of portfolio selection

- First Order Conditions: Derivative w.r.t weights set to zero.

$$2\Sigma w - \lambda\mu - \gamma\mathbf{1} = 0$$

- Result: The optimal weight vector w* is a closed-form solution.

$$w^* = \frac{1}{2}\Sigma^{-1}(\lambda\mu + \gamma\mathbf{1})$$

- Two-Fund Separation Theorem: The optimal portfolio is a linear mix of two 'funds':
  - 1. The purely return-seeking portfolio. $\Sigma^{-1}\mu$ (not really purely return-seeking) moves you on the efficient frontier towards better return (but at the lowest risk)
  - 2. The purely risk-minimizing portfolio. $\Sigma^{-1}1$ (nose of the bullet, the global minimum variance portfolio)
- No short constraints necessitate QP and Max Sharpe is non convex (Ratio of linear to quadratic root). Markowitz maximises variance for a target return. Max Sharpe finds the single 'Tangency Portfolio'.

$$\max \frac{w^T(\mu - r_f)}{\sqrt{w^T\Sigma w}} \quad \text{(Non-Convex)}$$

Final Solution in terms of μ_p

$$w^* = \mathbf{g} + \mathbf{h} \cdot \mu_p$$

This equation tells a simple story: Your optimal portfolio is just a **base portfolio** ($\mathbf{g}$) plus a **scaling factor** ($\mathbf{h}$) multiplied by your target return.

Where:

1. **The Intercept Vector ($\mathbf{g}$):**

$$\mathbf{g} = \frac{1}{\Delta}(C\Sigma^{-1}\mathbf{1} - B\Sigma^{-1}\mu)$$

(This is the "starting point" of the portfolio).

2. **The Slope Vector ($\mathbf{h}$):**

$$\mathbf{h} = \frac{1}{\Delta}(A\Sigma^{-1}\mu - B\Sigma^{-1}\mathbf{1})$$

(This is the direction you move in asset space as you demand more return).

3. **The Constants:**
   - $A = \mathbf{1}^T\Sigma^{-1}\mathbf{1}$
   - $B = \mathbf{1}^T\Sigma^{-1}\mu$
   - $C = \mu^T\Sigma^{-1}\mu$
   - $\Delta = AC - B^2$

Where does Markowitz fail?

- The optimizer treats your inputs (μ and Σ) as absolute truth. In reality, they are noisy, correlated estimates.
- **Information deficit problem:** To build a covariance matrix for $N = 500$ stocks, you must estimate 125,250 unique parameters (variances and covariances). However, 5 years of daily history only gives you ~1,250 data points. You are trying to solve a system with 100x more unknowns than equations.
- **Multicollinearity problem:** Because financial assets are highly correlated, they don't move independently. They move together in "clumps" (sectors/factors).

- **The Result:** Multicollinearity results in many stocks having correlations that are high (say 90% in reality). The noise (information deficit) results in creating some combinations where our estimate would say 99%. The optimizer thinks “Wow! Stock A and Stock B are almost identical, but Stock A has a slightly higher return in this sample! I have found a risk-free money printer!”. The optimizer exploits this "difference" by going **Long 1000% Stock A** and **Short 990% Stock B**

- **The Quote:** "Mean-variance optimization is an error-maximization machine." (Richard Michaud).

What are some alternate approaches to ex-post optimal portfolio using ML

- There are two broad approaches to solving this problem: Ex-post and Ex-Ante
  - Ex-post approaches Regularization (Ledoit Wolf), Hierarchical Risk Parity (HRP)
  - Ex-Ante approaches: forecast returns and provide expected returns to the optimizer (RFs, XGBoost, LSTMs, Transformers), Reinforcement Learning for end-to-end

Regularization

- We learned ridge regression in our ML 101 course

$\hat{\beta}^{ridge} = \min\{\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2\}$ (this is objective function to minimize)

The solution has a closed form, which can be written in matrix form as $\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$

- It basically shrinks the sample covariance towards the identity matrix. This implies an assumption that all stocks are uncorrelated and have equal variance. This is not really true.
- Step in the cousin of ridge: The Ledoit-Wolf. Here we fix the estimate of the covariance matrix itself. $\Sigma_{LW} = (1 - \delta)S + \delta F$
- The target F is rarely the identify matrix. It is usually the constant correlation matrix (assuming all stocks have the market correlation of around 0.5).
- How did we compute lambda in ridge? Ledoit and Wolf derived an optimal formula that minimizes MSE between estimated and true matrix

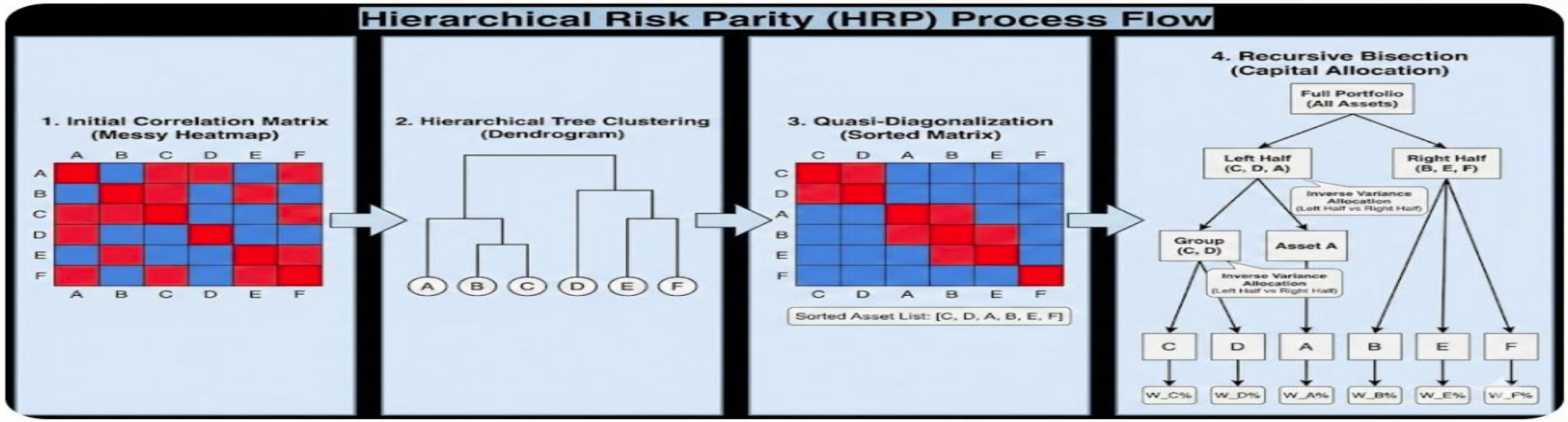
$$\delta^* = \frac{\sum_{i-1}^{N}\sum_{j-1}^{N} \mathrm{Var}(s_{ij})}{\sum_{i-1}^{N}\sum_{j-1}^{N}(f_{ij} - s_{ij})^2}$$

Hierarchical Risk Parity

- Hierarchical Risk Parity (HRP) does away with the need to invert a covariance matrix entirely. Instead, it uses machine learning techniques (hierarchical clustering) to build a robust portfolio that is far less sensitive to noise in the data, aiming for genuine diversification across different sources of risk.
- Risk Management algorithm not return maximizer. Comparable to minimum variance portfolio (MVP)
- However, MVP tries to find a global perfect portfolio through optimization which requires us to invert the covariance matrix which (thanks to information deficit and multicollinearity) results in haywire opposing weights that are merely the result of noise and don’t have any generalization.
- HRP is structurally forced to diversify in a top-down manner.

Hierarchical Risk Parity

- Steps:
  - Hierarchical cluster
  - Optional step: clustering based asset allocation
  - Quasi-diagonalisation
  - Recursive bisection

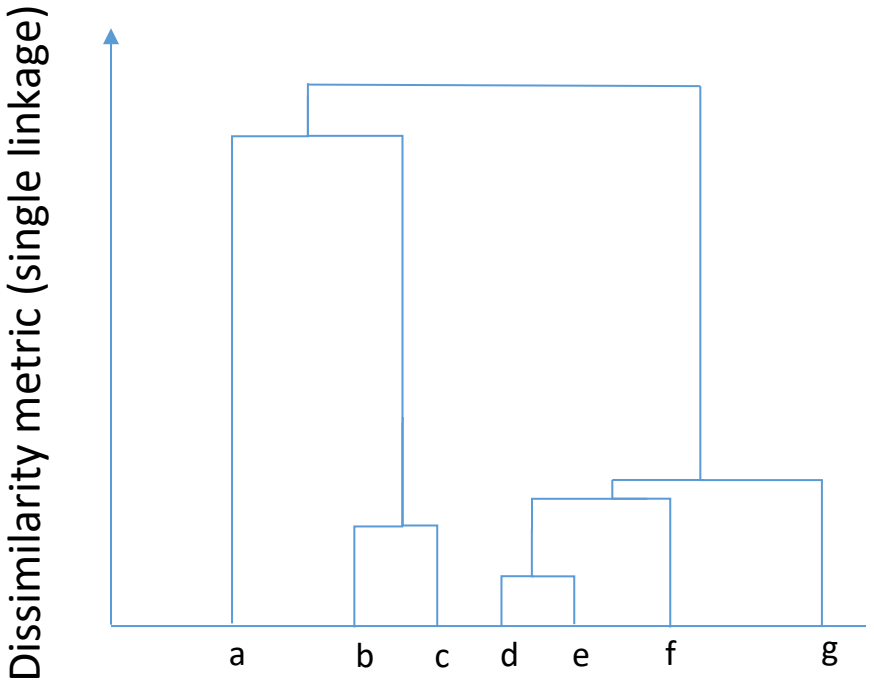


Hierarchical Risk Parity

- Hierarchical Clustering (quick revision)
  - Does not require a fixed number of clusters
  - Creates nested clusters
  - Types:
    - Agglomerative (Bottom-up)
    - Divisive (Top-down)
  - Requires defining some measure of dissimilarity between clusters
    - Single Linkage (why does HRP take single linkage?)
    - Complete Linkage
    - Group Average
    - Ward's method, increase in SSE from a merger

Hierarchical Risk Parity

Agglomerative Hierarchical Clustering



OLO ordering (a,b,c just for convenience)

Points to note:

- Graphical representation through Dendrogram
- Monotonicity
- What is the optimal number of clusters
- Optimal leaf ordering (OLO)
  - Hard constraint (nothing comes between a merger)
  - Soft constraint (order elements inside a cluster to be close to neigbhors). This soft constraint will play an important role in the quasi diagonal matrix

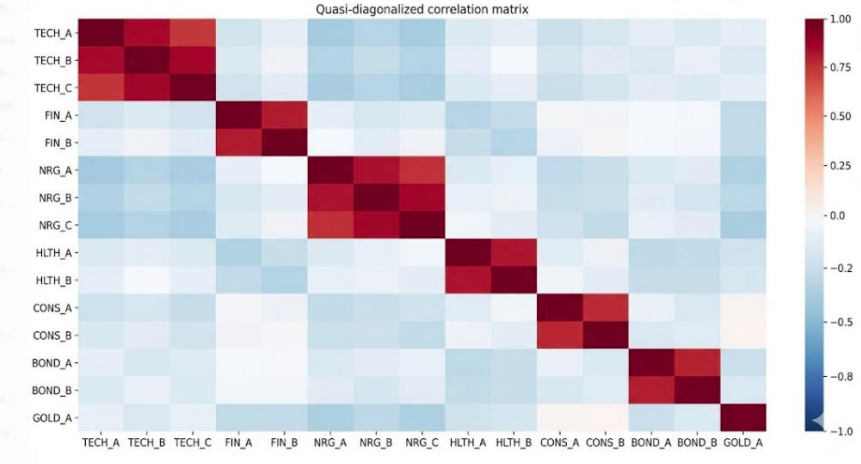
Why do Hierarchical clustering for HRP?:

- Hierarchical clustering-based asset allocation: replace clusters with single prototypical stocks
  - The stock with the highest correlation to the cluster or most liquid
- **We don't care about the clustering we care about the ordering!**

Hierarchical Risk Parity

- Quasi diagonalization – the sorted matrix is a visual verification of the clustering algorithms effect on ordered covariance.
  - We take the output from the dendrogram (a list of indices) and shuffle the columns of the covariance matrix to match the order.
- Recursive Bisection
  - Split the universe into two (left side vs right side)
  - Calculate cluster variance
    - Assign weights as per inverse variance parity
    - Combined cluster variance $V_{cluster} = w^T \cdot \Sigma_{cluster} \cdot w$
  - Calculate split factor

$$\alpha_{Left} = 1 - \frac{V_{Left}}{V_{Left} + V_{Right}}$$



  - Repeat the steps for next split till we have each stock weight. Why do we need quasi diagonlization for this to work?

What are some ex-ante approaches to optimal portfolios using ML

Ex-ante portfolio optimization focuses on constructing portfolios based on **forward-looking** expectations rather than solely relying on historical (ex-post) averages.

1. The "Predict-then-Optimize" Framework
    a. Feature Based Regression
    b. Time-Series Models
    c. Cross-Sectional Models

2. End-to-End Learning
    a. Deep Portfolios
    b. Reinforcement Learning

The problem of the replicating portfolio: Some ML approaches

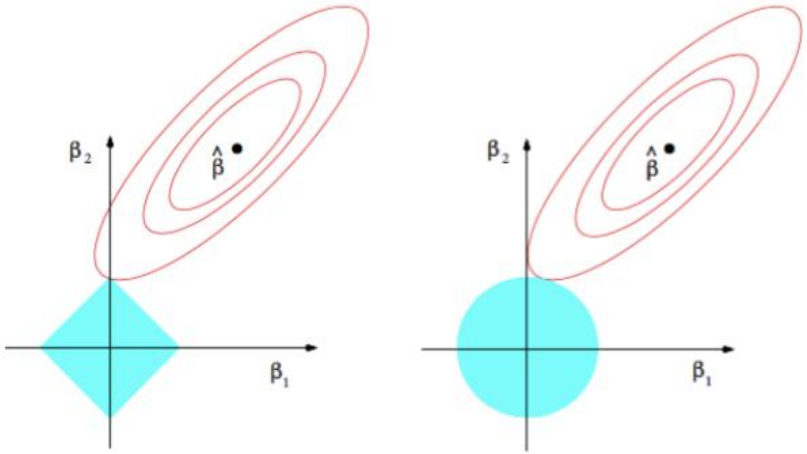
- What is the replicating portfolio problem
  - Standard MPT based optimization, L2 or Ledoit-Wolf regularization, HRP (without the optional step) will return weights for all the securities we send in for the portfolio construction
  - At its core the replicating portfolio problem is an exercise in signal compression. How much of market's essence can we capture with the smallest footprint? Why?
    - Cost efficiency
    - Market completeness and synthetic assets
    - Hedging and in the extreme case arbitrage
    - Mean reversion, Lead-Lag effects
- A few ML based approaches to the replicating portfolio problem
  - Lasso Regression
  - Autoencoders

Lasso Regression for the replicating portfolio

- Why is Lasso a natural choice the replicating portfolio problem

$$\hat{\beta}^{\text{lasso}} = \min\{\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} |\beta_j|\}$$

  - How is this different from Ridge?
  - The algebraic intuition for zero weights
  - What is the effect of diamond constraint vs circular constraint?

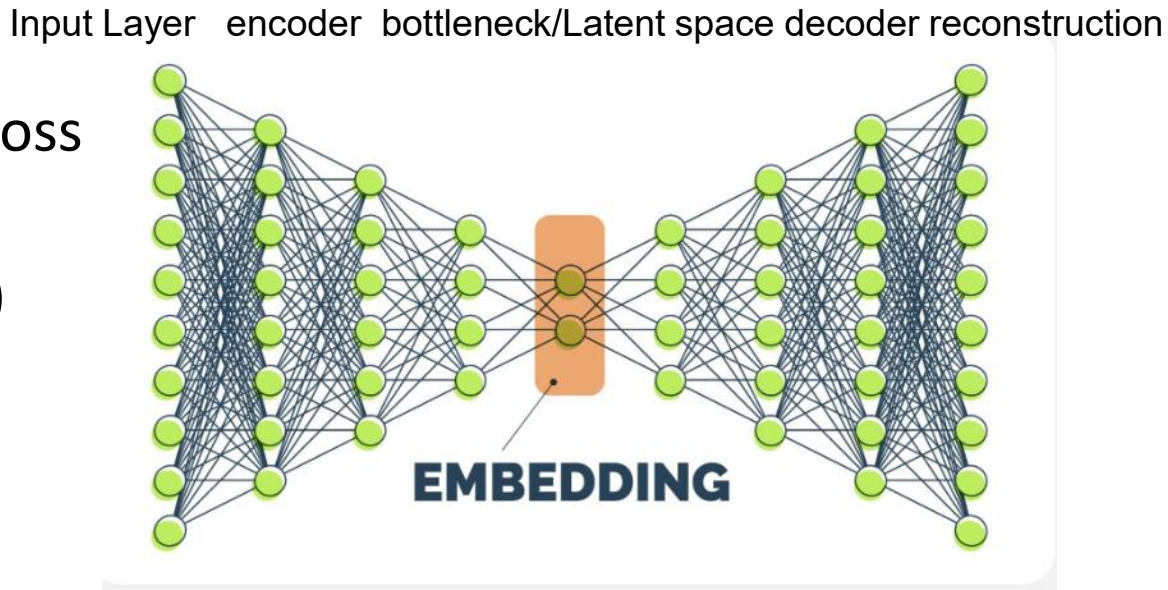


Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. "The elements of statistical learning." (2009).

Autoencoders

- Builds on the traditional approach of PCA--which is linear and does not allow for complex interaction effects.
- Deep Autoencoder approaches
  - Communality-based reconstruction Loss

  2016 Heaton, Polson, and Witte paper
  - Latent Factor Mapping (Hidden State)



Autoencoders

- Communality-based approaches
  - An autoencoder network minimizes reconstruction error $L(X, \hat{X}) = ||X - \hat{X}||^2$
  - We measure the communality of the stock (like R^2)

    $C_i = 1 - \frac{SSE_i}{SST_i}$ where $SSE_i = \sum_{t=1}^{T}(r_{t,i} - \hat{r}_{t,i})^2 = \sum_{t=1}^{T}(\epsilon_{t,i})^2$ and $SST_i = \sum_{t=1}^{T}(r_{t,i} - \bar{r}_i)^2$

  - We pick top M stocks with best communality
- Latent Factor Mapping
  - Get the ‘k’ synthetic factors of the middle node
  - For each synthetic factor pick the top ‘M/k’ stocks with the highest absolute correlation.
- Once you have ‘M’ stocks you can regress with the index, do MPT or HRP